

Smart Agriculture Irrigation System

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This is an Internet of Things (IoT) based irrigation system that automatically turns on water-pump when soil moisture in the field goes below the set threshold level. It has three parts—transmitter, receiver, and the IoT system—as may be seen from its block diagram in Fig. 1.

Circuit and working

Besides the IoT, the circuit comprises transmitter and receiver whose diagrams are shown in Fig. 2 and Fig. 3, respectively.

Transmitter. The transmitter is built around 433MHz RF transmitter module (TX1), encoder HT12E (IC1), and transistor BC547 (T1). It is installed in the field along with soil moisture sensor (SS1). The sensor's signal is transmitted by TX1 to the receiver, which is installed nearby—within a few metres from the transmitter. The transmitter works off 5V DC supply.

Threshold level of the soil moisture can be set using the in-built pot in soil sensor module. When soil moisture in the field goes below the threshold level, the sensor produces active low (0V) signal, which goes to the base of inverting transistor T1. Encoder IC1 receives active-high signal at pin 10 (AD8) from collector of T1 and generates serial data at pin 17 (DOUT) with its address pins A0 through A7 connected to ground. The modulated ASK RF signal from DOUT is transmitted by TX1 module at 433MHz frequency. J1 is used for optional sensor input for an optional motor in the receiver circuit.

Receiver. The receiver comprises 433MHz RF receiver module (RX1), decoder HT12D (IC2), two BC317 transistors (T2 and T3), a 5V single-changeover relay (RL1), and NodeMCU (Board1). This receiver also operates off a 5V DC supply.

The signal from transmitter is

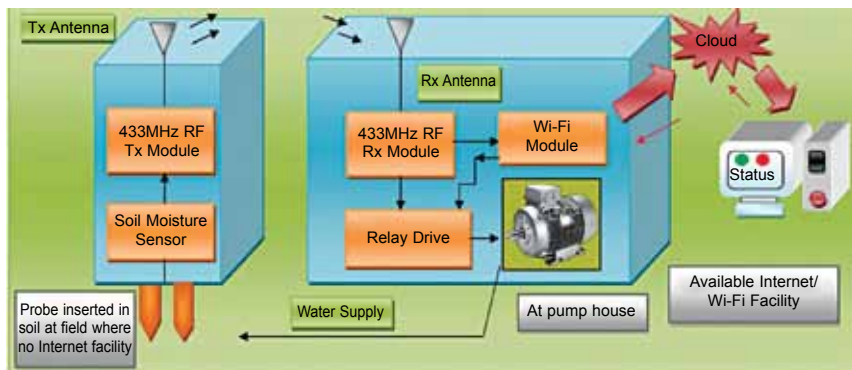


Fig. 1: Block diagram of the irrigation system

received by 433MHz receiver module RX1. Decoder HT12D, with address pins A0 through A7 connected to ground, receives the serial data at its pin 14 from RX1. Data pin D8 of IC2 drives the relay through transistors T2 and T3.

Diode 1N4007 (D2) connected in reverse bias mode is used to protect the circuit against voltage spikes during switching on/off of the water-pump. LED3 is for power on indication, LED4 for RF signal, and LED1 for water-pump's motor on status. If optional sensor is used at J1 in the transmitter, LED2 indicates the optional motor's status.

IoT system

The heart of the IoT system is NodeMCU ESP32/8266 Wi-Fi module. Data pin D8 of IC2 is connected to pin D0 of NodeMCU module (Board1) to provide the motor on/off status on a remote

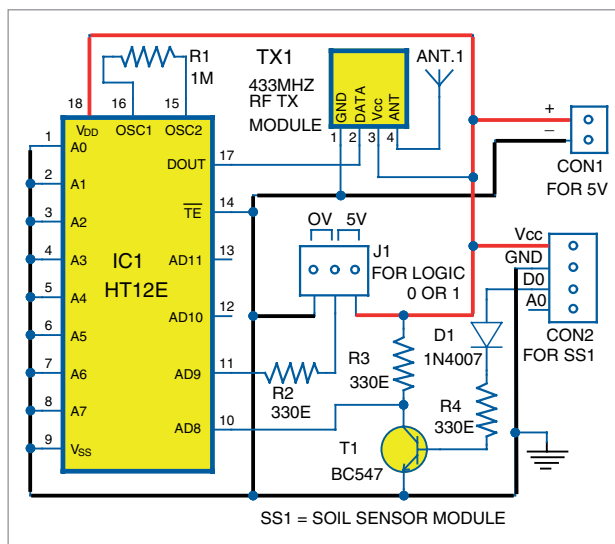


Fig. 2: Circuit diagram of transmitter

computer or smartphone over cloud. Adafruit IoT platform is used to monitor the water-pump's status online.

Pump's switching off. When soil is adequately wet, SS1 output pin D0 is low, which makes transistor T1 to stop conduction. So, collector voltage of T1 becomes high (5V), which is connected to pin AD8 of IC1. Transmitter TX1 transmits this signal, which is received by receiver RX1, decoded, and passed on to transistor T2 through resistor R10. So, T2 conducts making